Internet Appendix

for

Cybersecurity Risk

Table IA.1
Percentage of Firms with Cyber-related Disclosures by Industry and Year

This table presents the percentage of firms with cyber-related disclosures by (i) year and (ii) Fama and French 12 industry, where: 1- Consumer Non Durables; 2 -Consumer Durables; 3-Manufacturing; 4-Energy Oil and Gas; 5-Chemicals and Allied Products; 6-Business Equipment; 7-Telephone and Television Transmission; 8-Utilities; 9-Wholesale, Retail, and Some Services; 10-Healthcare, Medical Equipment, Drugs; 11-Money Finance and 12-Other.

	Full	*Fama-French Industry Group =*											
Year	Sample	1	2	3	4	5	6	7	8	9	10	11	12
2007	0.29	0.27	0.13	0.12	0.07	0.15	0.41	0.46	0.09	0.36	0.23	0.36	0.25
2008	0.31	0.26	0.11	0.13	0.07	0.22	0.42	0.54	0.10	0.39	0.23	0.41	0.27
2009	0.35	0.27	0.17	0.11	0.07	0.28	0.46	0.55	0.19	0.46	0.26	0.48	0.33
2010	0.39	0.36	0.17	0.15	0.08	0.32	0.49	0.57	0.26	0.53	0.28	0.53	0.36
2011	0.55	0.50	0.35	0.34	0.28	0.54	0.64	0.77	0.70	0.68	0.43	0.65	0.51
2012	0.66	0.60	0.49	0.49	0.42	0.60	0.75	0.81	0.83	0.78	0.52	0.74	0.61
2013	0.73	0.71	0.68	0.57	0.57	0.72	0.79	0.89	0.91	0.86	0.61	0.81	0.68
2014	0.80	0.79	0.80	0.71	0.65	0.75	0.84	0.93	0.94	0.90	0.70	0.86	0.78
2015	0.85	0.83	0.85	0.81	0.72	0.77	0.90	0.96	0.97	0.91	0.78	0.90	0.81
2016	0.88	0.86	0.89	0.85	0.78	0.83	0.91	0.96	0.97	0.94	0.84	0.89	0.86
2017	0.90	0.90	0.93	0.88	0.85	0.88	0.93	0.97	0.95	0.94	0.88	0.91	0.88
2018	0.89	0.95	0.92	0.87	0.60	1.00	0.95	1.00	1.00	0.90	0.82	0.86	0.82

Table IA.2
Cybersecurity Risk and Firm Characteristics-Jaccard Similarity

This table reports the results of linear regressions of firm characteristics on cybersecurity risk as measured by Jaccard similarity. All variables are defined in Appendix B. Standard errors are clustered at the firm level. *, **, and *** indicate significance at the 10%, 5% and 1% levels, respectively.

	Model 1		Model 2	
Firm Size (ln)	**0.006**	***	**0.007**	***
	[14.44]		[4.94]	
Firm Age (ln)	**-0.003**	***	**-0.016**	***
	[-4.72]		[-5.73]	
Tobin's Q	**0.004**	***	**0.001**	*
	[7.84]		[1.89]	
ROA	**0.021**	***	**0.012**	***
	[5.55]		[3.25]	
Tanginility	**-0.034**	***	**-0.010**	
	[-8.78]		[-1.28]	
R&D Expenditures	**-0.008**		**0.037**	***
	[-0.99]		[4.72]	
Secrets	**0.010**	***	**0.013**	***
	[5.91]		[4.70]	
Cash Flow Volatility (Industry)	**-0.095**	***	**0.006**	
	[-6.70]		[0.42]	
Risk Section Length (ln)	**0.022**	***	**0.020**	***
	[39.70]		[20.15]	
Readability (ln)	**0.002**	**	**0.002**	*
	[2.36]		[1.73]	
Institutional Ownership	**0.010**	***	**0.006**	
	[2.66]		[1.57]	
Independent Directors	**0.037**	***	**0.015**	*
	[5.04]		[1.68]	
Risk Committee	**0.007**	**	**0.000**	
	[2.51]		[-0.10]	
Constant	**-0.175**	***	**-0.112**	***
	[-12.03]		[-6.33]	
No of Observations	35,308		35,308	
Clustered SE	Firm		Firm	
Firm fixed effects	No		Yes	
Industry fixed effects	Yes		No	
Year fixed effects	Yes		Yes	
R-Squared	0.510		0.782	

Table IA.3

Cybersecurity Risk and Negative Asymmetries in Stock Returns- Jaccard Similarity

This table reports the results of regressions of cybersecurity risk on two different proxies for negative asymmetries in stock returns. In Model 1, we use *NCSKEW*, which equals the negative of the third moment of firm-specific weekly returns for each firm in a year divided by the standard deviation of firm-specific weekly returns raised to the third power. In Model 2, we use *EXTR_SIGMA*, which is the negative of the worst deviation of firm-specific weekly returns from the average firm specific weekly return divided by the standard deviation of firm-specific weekly returns. Cybersecurity risk is measured at the beginning of each year using Jaccard similarity. All variables are defined in Appendix B. *, **, and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively.

	NCSKEW	*EXTR_SIGMA*
	Model 1	Model 2
Cybersecurity Risk Index (Jaccard)	**0.312** ***	**0.280** ***
	[3.53]	[3.47]
Firm Size (ln)	**0.047** ***	**0.025** ***
	[5.03]	[2.98]
Firm Age (ln)	**-0.030** ***	**-0.023** ***
	[-3.94]	[-3.25]
Tobin's Q	**-0.086** ***	**-0.075** ***
	[-9.65]	[-10.11]
ROA	**0.022**	**0.036** ***
	[1.49]	[2.70]
Tanginility	**-0.020** **	**-0.032** ***
	[-2.24]	[-4.00]
R&D Expenditures	**0.045** ***	**0.053** ***
	[2.98]	[3.78]
Secrets	**0.019** ***	**0.022** ***
	[2.63]	[3.26]
Cash Flow Volatility (Industry)	**0.004**	**0.002**
	[0.40]	[0.23]
Risk Section Length (ln)	**0.016** ***	**0.009**
	[2.61]	[1.59]
Readability (ln)	**-0.015** *	**-0.010**
	[-1.88]	[-1.38]
Institutional Ownership	**0.043** ***	**0.029** ***
	[6.85]	[5.07]
Independent Directors	**-0.013** *	**-0.007**
	[-1.95]	[-1.08]
Risk Committee	**-0.033**	**-0.050** **
	[-1.58]	[-2.38]
Constant	**0.211** ***	**2.714** ***
	[9.05]	[116.4]
Clustered SE	Firm	Firm
Industry fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Number of Observations	24,657	24,657
R-squared	0.025	0.029

Table IA.4
Cybersecurity Risk and Future Cyberattacks-Jaccard Similarity

This table reports the results of logit regressions of cybersecurity risk (Jaccard similarity) on future cyberattacks. Panel A includes all cyberattacks reported in PRC database for which we have complete risk disclosure and financial data. In Panel B we restrict our attention to major cyberattacks and in particular those that attracted attention by global news outlets (e.g. CNBC, Financial Times and the Wall Street Journal) and covered in major Newswires (e.g. AP, Bloomberg, Reuters). In Panel C we restrict our attention to non-major cyberattacks (those that did not attract attention from major Newswires). Future cyberattacks are measured at time *t+1* while all independent variables are measured at time *t*. All variables are defined in Appendix B. Standard errors are clustered at the firm level. *, **, and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively.

	Panel A: All Cyber Attacks		*Panel B: Major Cyber Attacks*		*Panel C: Non-major Cyber Attacks*	
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Cybersecurity Risk Index (Jaccard)	**0.749** ***	**0.504** ***	**0.743** ***	**0.562** ***	**0.721** ***	**0.440** ***
	[8.67]	[5.37]	[5.45]	[4.75]	[8.20]	[3.65]
Previous Attack Dummy	-	**1.435** ***	-	**1.622** ***	-	**1.061** **
	-	[3.76]	-	[2.99]	-	[2.14]
Firm Size (ln)	-	**1.496** ***	-	**1.814** ***	-	**1.236** ***
	-	[10.55]	-	[8.36]	-	[7.61]
Firm Age (ln)	-	**-0.121**	-	**-0.221**	-	**-0.031**
	-	[-1.10]	-	[-1.38]	-	[-0.22]
Tobin's Q	-	**0.190**	-	**0.300**	-	**0.087**
	-	[1.28]	-	[1.57]	-	[0.43]
ROA	-	**0.505**	-	**0.416**	-	**0.597**
	-	[1.59]	-	[1.11]	-	[1.39]
Tanginility	-	**-0.038**	-	**-0.192**	-	**0.065**
	-	[-0.26]	-	[-0.86]	-	[0.36]
R&D Expenditures	-	**-0.016**	-	**-0.149**	-	**0.068**
	-	[-0.04]	-	[-0.31]	-	[0.13]
Secrets	-	**0.280** ***	-	**0.105**	-	**0.393** ***
	-	[2.87]	-	[0.75]	-	[3.08]
Cash Flow Volatility (Industry)	-	**-0.177**	-	**-0.531**	-	**-0.029**
	-	[-0.79]	-	[-1.43]	-	[-0.10]
Risk Section Length (ln)	-	**-0.185**	-	**-0.389** **	-	**0.017**
	-	[-1.42]	-	[-2.24]	-	[0.12]
Readability (ln)	-	**0.017**	-	**-0.102**	-	**0.121**
	-	[0.12]	-	[-0.49]	-	[0.61]
Institutional Ownership	-	**0.139**	-	**0.444** ***	-	**-0.082**
	-	[1.17]	-	[2.70]	-	[-0.57]
Independent Directors	-	**-0.067**	-	**-0.158**	-	**0.012**
	-	[-0.59]	-	[-1.11]	-	[0.07]
Risk Committee	-	**-0.203**	-	**-0.448**	-	**-0.037**
	-	[-0.49]	-	[-1.02]	-	[-0.06]
Constant	**-8.035** ***	**-8.607** ***	**-8.594** ***	**-9.867** ***	**-8.783** ***	**-9.027** ***
	[-10.12]	[-10.71]	[-7.80]	[-9.13]	[-8.04]	[-8.06]
Clustered SE	Firm	Firm	Firm	Firm	Firm	Firm
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Number of Observations	41,140	30,830	38,934	30,830	41,140	30,830
Pseudo-R-squared	0.094	0.223	0.090	0.244	0.086	0.196

Table IA.5
Cybersecurity Risk Portfolios-Jaccard Similarity

This table reports average excess returns, CAPM alphas, four-factor alphas from Carhart's (1997) FFC model (FFC alphas) and five-factor alphas from Fama and French's (2015) model (Five-Factor alphas) for portfolios constructed on the basis of our Cybersecurity Risk Index, as measured by Jaccard similarity. Starting from December 2007, we sort stocks at the end of each quarter in ascending order on the basis of their Cybersecurity Risk and allocate them into three groups (Low Cyber-Risk Stocks, Middle Group and High Cyber-Risk Stocks). We track the performance of the three portfolios over the following quarter until these are rebalanced. We form the spread strategy P3-P1 that is long the portfolio with the highest cybersecurity-risk stocks (P3) and short the portfolio with the lowest cybersecurity-risk stocks (P1). Returns are reported for equally-weighted (ew) and value-weighted (vw) portfolios over the period March 2008- March 2019. Average (monthly) excess portfolio returns and alphas are bolded; their associated Newey-West *t*-statistics are reported in square brackets. We exclude from the analysis firms that appear in a sample for a period less than 3 years and have zero disclosures on cyber-related issues throughout that period. *, ** and *** denote statistical significance at 10%, 5% and 10% levels, respectively.

Future (1-month) portfolio returns sorted by our Cybersecurity Risk Index					
		Portfolios			
		Low Cyber-Risk	*Middle Group*	*High Cyber-Risk*	
		[P1]	[P2]	[P3]	[P3]-[P1]
Excess return	ew	**0.169**	**0.701** *	**0.852** **	**0.683** ***
		[0.38]	[1.71]	[2.15]	[4.70]
	vw	**0.508**	**0.883** ***	**1.025** ***	**0.517** **
		[1.25]	[2.62]	[2.95]	[2.33]
CAPM alpha	ew	**-0.727** **	**-0.214**	**-0.059**	**0.668** ***
		[-3.32]	[-0.98]	[-0.39]	[4.71]
	vw	**-0.339** *	**0.090**	**0.181** *	**0.520** **
		[-1.90]	[0.80]	[1.85]	[2.37]
FFC alpha	ew	**-0.675** ***	**-0.154**	**-0.005**	**0.670** ***
		[-4.87]	[-1.41]	[-0.06]	[4.70]
	vw	**-0.277** *	**0.103**	**0.166**	**0.443** **
		[-1.87]	[1.04]	[1.61]	[2.26]
Five-factor alpha	ew	**-0.602** ***	**-0.111**	**0.058**	**0.660** ***
		[-3.80]	[-0.79]	[0.69]	[4.40]
	vw	**-0.306** **	**0.053**	**0.186**	**0.492** ***
		[-2.30]	[0.49]	[1.55]	[2.58]

Table IA.6
Cross-sectional Fama-MacBeth Regressions: Jaccard Similarity

This table reports the results from Fama-MacBeth regressions on the relation between our Cybersecurity Risk Index, as measured Jaccard similarity and subsequent stock returns (1-month). For each month of our sample we run cross-sectional regressions of excess stock returns on lagged cybersecurity risk and a set of firm characteristics that are also lagged. These include beta, size, book-to-market, momentum, short-term reversal, illiquidity, coskewness, idiosyncratic volatility, asset growth, profitability, R&E expenditures, demand for lottery-like stocks (max), length of Item 1A. Risk Factors of the Form 10-K and 10-K readability. All variables are defined in Appendix B. The continuous variables are standardized to have a mean of 0 and standard deviation of 1. The coefficients are reported as time-series averages of the estimates from the cross-sectional regressions. The *t*-statistics, which are reported in brackets, are based on the Newey-West heteroskedasticity and autocorrelation consistent standard errors. *, ** and *** denote statistical significance at 10%, 5% and 1% levels, respectively.

	$Returns_{t+1}$		
	[1]	[2]	[2]
Cybersecurity Risk Index (Jaccard)	**0.297** ***	**0.111** ***	**0.136** ***
	[6.34]	[2.83]	[3.03]
Beta	-	**0.089**	**0.091**
		[0.82]	[0.86]
Market Value	-	**-0.086**	**-0.061**
		[-1.28]	[-0.83]
Book-to-Market	-	**0.068**	**0.068**
		[1.27]	[1.34]
Momentum	-	**0.153** *	**0.147** *
		[1.75]	[1.69]
Reversal	-	**-0.116** **	**-0.122** **
		[-2.07]	[-2.21]
Illiquidity	-	**-0.013**	**-0.015**
		[-0.35]	[-0.42]
CoSkew	-	**-0.029**	**-0.026**
		[-0.95]	[-0.85]
Indiosyncratic Volatility	-	**-0.476** ***	**-0.468** ***
		[-5.78]	[-5.80]
Asset Growth	-	**-0.109** ***	**-0.100** **
		[-2.76]	[-2.65]
ROA	-	**0.519** ***	**0.504** ***
		[7.61]	[8.10]
R&D Expenditures	-	**0.321** ***	**0.318** ***
		[4.44]	[5.08]
Max	-	**-0.423** ***	**-0.416** ***
		[-4.58]	[-4.57]
Risk Section Length (ln)	-	-	**-0.062**
			[-1.62]
Readability (ln)	-	-	**-0.033**
			[-0.64]
Constant	**0.515**	**0.512**	**0.509**
	[1.14]	[1.12]	[1.12]
Observations	409,016	342,573	342,573

Table IA.7
Cybersecurity Risk Portfolios-Robustness Tests

This table reports average excess returns and alphas from the Fama and French's (2015) model (Five-Factor alphas) for the spread strategy that is long the portfolio with the highest cybersecurity-risk stocks and short the portfolio with the lowest cybersecurity-risk stocks. Results are reported both for equally-weighted and value-weighted portfolios. We exclude from the analysis firms that appear in a sample for a period less than 3 years and have zero disclosures on cyber-related issues throughout that period. In Panel A we repeat our portfolio analysis for the period January 2012-March 2019 (Post SEC's guidance on cybersecurity). In Panel B, we form our portfolios based on another revised cybersecurity risk measure, which replaces all zeros with the industry/sector median value in any given year. In Panel C, we form our portfolios based on a revised cybersecurity risk measure, which replaces all zeros with the next non-zero observation for each firm. In Panel D, we exclude from the portfolio analysis all firms that belong to the same Fama-French 48 industry as firms in the training sample (i.e., peer firms). In Panel E, we form the portfolios based on a new cybersecurity risk measure, which is constructed after estimating the similarity of firm *i*'s cybersecurity-risk disclosure with past cybersecurity-risk disclosures of firms in a new training sample that *excludes* firms that have the same auditor as firm *i*. In Panel F, we repeat our portfolio analysis using industry-adjusted returns. To adjust returns by industry, we focus on the Fama-French 12 industry portfolios and their monthly average returns. For robustness purposes, we use both equal-weighted and value-weighted industry-level returns for the industry adjustment. In Panel G, we repeat the analysis 12 times after excluding each of the Fama-French 12 industries in turn to flush out abnormal impact of any particular industry group. In Panel H, we report portfolio results after excluding firms from the Energy Oil and Gas and Consumer Durables industries. In Panel I (Panel J) we exclude from the analysis all firms with cyber insurance (firms in the training sample). In Panel K, we report results after forward-filling our measure and extending the sample to December 2020. Finally, in In Panel L we present results based on monthly and yearly rebalancing. Average (monthly) excess portfolio returns and alphas are bolded; their associated Newey-West t-statistics are reported in square brackets. *, ** and *** denote statistical significance at 10%, 5% and 1% levels, respectively.

	Equal-weighted portfolios High - Low Cyber Risk Stocks		Value-weighted portfolios High - Low Cyber Risk Stocks	
	Avg. Return	*5-Factor alpha*	*Avg. Return*	*5-Factor alpha*
Panel A: Post SEC's Guidance Period				
January 2012 to March 2019	**0.916 *****	**0.870 *****	**0.770 *****	**0.652 *****
	[5.88]	[5.56]	[3.00]	[3.45]
Panel B: Replacing zeros with industry medians				
Tercile Portfolios (P3-P1)	**0.572 *****	**0.660 *****	**0.369 ** **	**0.470 *****
	[3.87]	[5.09]	[2.07]	[2.62]
Quartile Portfolios (P4-P1)	**0.601 *****	**0.723 *****	**0.393 * **	**0.569 *****
	[3.27]	[4.56]	[1.91]	[3.40]
Quintile Portfolios (P5-P1)	**0.602 *****	**0.704 *****	**0.459 ** **	**0.584 *****
	[3.04]	[4.10]	[2.07]	[3.27]
Decile Portfolios (P10-P1)	**0.527 ** **	**0.673 *****	**0.387**	**0.487 ** **
	[2.43]	[3.80]	[1.49]	[2.36]
Panel C: Replacing zeros with next non-zero obs.				
Tercile Portfolios (P3-P1)	**0.235 ** **	**0.300 *****	**0.289 *****	**0.289 *****
	[2.08]	[2.79]	[3.11]	[2.73]
Quartile Portfolios (P4-P1)	**0.306 ** **	**0.386 *****	**0.283 ** **	**0.308 ** **
	[2.27]	[3.17]	[2.49]	[2.23]
Quintile Portfolios (P5-P1)	**0.384 *****	**0.471 *****	**0.314 ** **	**0.363 ** **
	[2.66]	[3.61]	[2.16]	[2.35]
Decile Portfolios (P10-P1)	**0.515 *****	**0.423 ** **	**0.631 *****	**0.529 ** **
	[2.75]	[2.24]	[3.51]	[2.46]
Panel D: Cybersecurity Risk after:				
Excluding peer firms (those that belong to the same FF48 industry with training firms)	**0.709 *****	**0.676 *****	**0.638 *****	**0.631 *****
	[4.57]	[4.19]	[3.12]	[3.69]
Panel E: Cybersecurity Risk after:				
Excluding peer firms (same Auditor) from training sample	**0.757 *****	**0.748 *****	**0.553 *****	**0.540 *****
	[5.54]	[5.40]	[2.64]	[3.16]

Table Continued Overleaf

Table IA.7 (continued)

	Equal-weighted portfolios High - Low Cyber Risk Stocks		Value-weighted portfolios High - Low Cyber Risk Stocks	
	Avg. Return	*5-Factor alpha*	*Avg. Return*	*5-Factor alpha*
Panel F: Industry Adjustment based on				
Equal-Weighted Industry-Level Returns	**0.640** ***	**0.653** ***	**0.541** ***	**0.569** ***
	[5.82]	[5.53]	[2.60]	[3.14]
Value-Weighted Industry-Level Returns	**0.618** ***	**0.601** ***	**0.489** ***	**0.484** ***
	[4.65]	[4.50]	[2.57]	[2.83]
Panel G: All Firms Excluding:				
Consumer Non_Durables	**0.704** ***	**0.675** ***	**0.698** ***	**0.644** ***
	[4.45]	[3.24]	[4.44]	[3.89]
Consumer Durables	**0.674** ***	**0.619** ***	**0.646** ***	**0.577** ***
	[4.63]	[3.07]	[4.42]	[3.57]
Manufacturing	**0.690** ***	**0.633** ***	**0.636** ***	**0.564** ***
	[5.01]	[3.22]	[4.28]	[3.44]
Energy Oil and Gas	**0.614** ***	**0.566** ***	**0.570** ***	**0.495** ***
	[4.60]	[2.80]	[3.99]	[2.88]
Chemicals and Allied Products	**0.666** ***	**0.638** ***	**0.647** ***	**0.592** ***
	[4.58]	[3.06]	[4.42]	[3.54]
Business Equipment	**0.575** ***	**0.519** **	**0.570** ***	**0.580** ***
	[3.56]	[2.46]	[3.52]	[3.27]
Telephone and Television Transmission	**0.685** ***	**0.609** ***	**0.666** ***	**0.571** ***
	[4.59]	[2.96]	[4.45]	[3.41]
Utilities	**0.688** ***	**0.620** ***	**0.672** ***	**0.585** ***
	[4.58]	[3.03]	[4.46]	[3.67]
Wholesale, Retail, and Some Services	**0.760** ***	**0.573** ***	**0.775** ***	**0.541** ***
	[4.54]	[2.91]	[4.92]	[3.44]
Healthcare, Medical Equipment, Drugs	**0.659** ***	**0.681** ***	**0.660** ***	**0.650** ***
	[4.31]	[3.17]	[4.47]	[3.75]
Money Finance	**0.746** ***	**0.711** ***	**0.671** ***	**0.692** ***
	[4.93]	[2.74]	[4.41]	[3.53]
Other	**0.665** ***	**0.450** ***	**0.661** ***	**0.409** ***
	[4.14]	[3.08]	[3.97]	[3.36]
Panel H: All Firms Excluding:				
Consumer Durables & Energy Oil and Gas	**0.601** ***	**0.574** ***	**0.551** ***	**0.499** ***
	[4.85]	[2.81]	[4.09]	[2.89]
Panel I: All Firms Excluding:				
Firms with Cyber Insurance	**0.660** ***	**0.634** ***	**0.676** ***	**0.649** ***
	[4.40]	[4.27]	[3.34]	[4.05]
Panel J: All Firms Excluding:				
Firms in Training Sample	**0.669** ***	**0.682** ***	**0.522** **	**0.507** **
	[4.42]	[4.67]	[2.20]	[2.42]
Panel K: Extended Sample				
March 2008 to December 2020	**0.737** ***	**0.679** ***	**0.747** ***	**0.623** ***
	[5.10]	[4.72]	[3.27]	[3.99]
Panel L: Alternative Rebalancing				
Monthly Rebalancing	**0.667** ***	**0.646** ***	**0.599** ***	**0.561** ***
	[4.48]	[4.32]	[2.95]	[3.49]
Yearly Rebalancing	**0.691** ***	**0.669** ***	**0.586** ***	**0.559** ***
	[4.73]	[4.52]	[2.84]	[3.40]

Table IA.8
Cybersecurity Risk and Future Cyberattacks: Dealing with Peer Effects in Disclosure Language

This table reports the results of logit regressions of cybersecurity risk (cosine similarity) on future cyberattacks. The dependent variable is a dummy variable that equals 1 if the firm experiences a cyberattack at time *t+1*, and zero otherwise. The key explanatory variable in Model 1 is our cybersecurity risk measure (benchmark specification-see section 3.5). In Model 2, we replace cybersecurity risk measure with the variable Cybersecurity Risk Index (Excluding Peers-Industry), which is simply the cosine similarity of firms that do not belong to the same Fama-French 48 industry as firms in the training sample (i.e., peer firms). In Model 3, we use the variable Cybersecurity Risk Index (Excluding Peers-Auditor) as our key explanatory variable. This is constructed after estimating the cosine similarity of firm's *i* cybersecurity-risk disclosure with past cybersecurity-risk disclosures of firms in a new training sample, that excludes firms that have the same auditor with firm *i*. All independent variables are measured at time *t*. The continuous variables are standardized to have a mean of 0 and standard deviation of 1. Standard errors are clustered at the firm level. *** denotes statistical significance at the 1% level.

	Model 1	Model 2	Model 3
Cybersecurity Risk Index	**0.656 *****	-	-
	[4.60]	-	-
Cybersecurity Risk Index (Excluding Peers-Industry)		**0.660 *****	
		[4.21]	
Cybersecurity Risk Index (Excluding Peers-Auditor)	-	-	**0.631 *****
	-	-	[4.77]
Clustered SE	Firm	Firm	Firm
Industry fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Control Variables	Yes	Yes	Yes
Number of Observations	30,830	25,280	30,059
Pseudo-R-squared	0.223	0.244	0.215

Table IA.9
Double-Sorting: R&D and Innovation Activity

This table reports average returns and 5-factor alphas from the Fama and French's (2015) model for double-sorted portfolios on the basis of the cybersecurity risk index and each of the following firm characteristics: (i) R&D Expenditures, defined as the ratio of R&D expenditures to total assets, (ii) Patent Flow, defined as the number of patents a firm produces in a given year and (iii) Patent Stock, defined as the number of patents a firm owns prior to, and up to, a given year. Data on patent stock and patent flow are drawn from the Duke Innovation & Scientific Enterprises Research Network (DISCERN*;* *https://doi.org/10.5281/zenodo.3594743*) database by Arora, Belenzon, and Sheer (2021a, 2021b). These data cover the period 1980 to 2015. For the purposes of our analysis, we forward-fill the missing data for years 2016-2018 with the 2015 values for the variables of interest. Starting from December 2007, we sort stocks at the end of each quarter in ascending order on the basis of their Cybersecurity Risk and allocate them into three groups (Low Cyber-Risk Stocks, Middle Group and High Cyber-Risk Stocks), and we also independently sort stocks into ascending order according to the value of each characteristic mentioned above and allocate them into two portfolios (LOW and HIGH) based on median values for each quarter. The intersection of these two classifications yields the double-sorted portfolios. We track the performance of the intersection portfolios over the following quarter until these are rebalanced. We report both equal-weighted and value-weighted average returns and five-factor alphas for the spread strategy High-Low Cyber Risk Stocks within each HIGH and LOW classification. Newey-West *t*-statistics are reported in square brackets. *, ** and *** denote statistical significance at 10%, 5% and 1% levels, respectively.

		Equal-weighted portfolios High - Low Cyber Risk Stocks		Value-weighted portfolios High - Low Cyber Risk Stocks	
		Avg. Return	*5-Factor alpha*	*Avg. Return*	*5-Factor alpha*
R&D Expenditures	LOW	**0.452** ***	**0.438** ***	**0.437** ***	**0.497** ***
		[2.92]	[2.72]	[2.85]	[3.62]
	HIGH	**0.986** ***	**0.924** ***	**0.830** **	**0.681** **
		[5.92]	[5.56]	[1.98]	[2.05]
Patent Flow	LOW	**0.604** ***	**0.502** ***	**0.189**	**0.317** **
		[4.63]	[3.57]	[1.11]	[2.16]
	HIGH	**0.831** ***	**0.802** ***	**0.930** *	**0.762** **
		[4.33]	[4.64]	[1.93]	[2.08]
Patent Stock	LOW	**0.688** ***	**0.590** ***	**0.232**	**0.351** *
		[5.09]	[4.09]	[1.58]	[1.95]
	HIGH	**0.748** ***	**0.712** ***	**0.879** *	**0.725** **
		[4.22]	[5.26]	[1.94]	[2.10]

Table IA.10
Placebo Tests

This table reports results after using a placebo measure that is based on similarity in risk disclosures, other than cyber-related disclosures (see section 6.2 for details). Panel A reports the results of logit regressions. The dependent variable is a dummy variable that equals 1 if the firm experiences a cyberattack at time *t+1*, and zero otherwise. The key variable of interest is the placebo measure (Non-Cyber Disclosure Similarity). Model 1 (Model 2) excludes from (includes in) the model our cybersecurity risk measure. Both models include standard controls (as in Table 6 of the paper), which are measured at time *t*. The continuous variables are standardized to have a mean of 0 and standard deviation of 1. Standard errors are clustered at the firm level. Panel B reports alphas from the Fama and French's (2015) model for the spread strategy that is long the portfolio of stocks with the highest similarity in overall risk disclosures (Non-Cyber Disclosure Similarity) and short the portfolio of stocks with the lowest similarity in overfall risk disclosures (Non-Cyber Disclosure Similarity). Results are reported both for equally-weighted and value-weighted tercile, quartile, quintile and decile portfolios. Average alphas are bolded and their associated Newey-West t-statistics are reported in square brackets. * and *** denote statistical significance at 10% and 1% levels, respectively.

Panel A: Future Attacks	Model 1	Model 2
Cybersecurity Risk Index	-	**0.727 *****
	-	[4.72]
Non-Cyber Disclosure Similarity	**-0.058**	**-0.268**
	[-0.31]	[-1.40]
Clustered SE	Firm	Firm
Industry fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Control Variables	Yes	Yes
Number of Observations	30,665	30,665
Pseudo-R-squared	0.201	0.225

Panel B: Portfolio Analysis	5-Factor Alpha of Spread Portfolios sorted on Non-Cyber Disclosure Similarity [High-Low]	
	Equal-weighted	*Value-weighted*
Tercile Portfolios (P3-P1)	**0.099**	**0.257**
	[0.92]	[1.63]
Quartile Portfolios (P4-P1)	**0.104**	**0.341 ***
	[0.84]	[1.73]
Quintile Portfolios (P5-P1)	**0.111**	**0.329 ***
	[0.86]	[1.71]
Decile Portfolios (P10-P1)	**-0.074**	**0.222**
	[-0.53]	[1.53]

Table IA.11
Fama-MacBeth Regressions: Controlling for Extra Risks

This table reports the results from Fama-MacBeth regressions on the relation between our Cybersecurity Risk Index and subsequent monthly stock returns (1-month). For each month of our sample we run cross-sectional regressions of excess stock returns on lagged cybersecurity risk and a set variables that capture other types of risk such as political, non-political and overall risk (Hassan et al., 2019) and climate risk (Sautner et al., 2020). These extra risk measures have been developed using textual analysis of earnings conference calls and the data are available online (see https://www.firmlevelrisk.com/ and https://osf.io/fd6jq/). The continuous variables are standardized to have a mean of 0 and standard deviation of 1. The coefficients are reported as time-series averages of the estimates from the cross-sectional regressions. The *t*-statistics, which are reported in brackets, are based on the Newey-West heteroskedasticity and autocorrelation consistent standard errors. *** denotes statistical significance at the 1%.

	$Returns_{t+1}$			
	[1]	[2]	[3]	[4]
Cybersecurity Risk Index	**0.217** ***	**0.216** ***	**0.216** ***	**0.226** ***
	[4.08]	[4.10]	[4.10]	[4.37]
Political Risk (Hassan et al., 2019)	**-0.007**	-	-	-
	[-0.23]			
Non-political Risk (Hassan et al., 2019)	-	**0.017**	-	-
		[0.58]		
Overall Risk (Hassan et al., 2019)	-	-	**0.002**	-
			[0.05]	
Climate Risk (Sautner et al., 2020)	-	-	-	**-0.048**
				[-1.41]
Observations	291,625	291,625	291,625	308,375

Table IA.12
Fama-MacBeth Regressions: Further Results

This table presents further evidence from Fama-MacBeth regressions on the relation between our Cybersecurity Risk Index and subsequent monthly stock returns (1-month). Panel A presents the correlation coefficients and corresponding levels of statistical significance of the relationship between our cybersecurity risk measure and the variables idiosyncratic volatility (IVOL) and ROA. In Panel B, we present results from Fama-Macbeth regressions that do not include (Model 1) and do include IVOL/ROA as controls (Model 2/ Model 3). In Panel C, we present results from Fama-Macbeth regressions after orthogonalizing our cybersecurity risk with respect to IVOL and also with respect to ROA. To do so, we regress our measure on IVOL and obtain the residuals. The residual provides us with a new cybersecurity risk measure that is orthogonal to IVOL, namely Cybersecurity Risk (Orthogonal to IVOL). We repeat this exercise by regressing cybersecurity risk on ROA to obtain another cybersecurity risk measure that is orthogonal to ROA, namely Cybersecurity Risk (Orthogonal to ROA). All variables are defined in Appendix B. The coefficients are reported as time-series averages of the estimates from the cross-sectional regressions. The *t*-statistics, which are reported in brackets, are based on the Newey-West heteroskedasticity and autocorrelation consistent standard errors. ** and *** denote statistical significance at 5% and 1%, respectively.

Panel A: Correlations	(i)	(ii)	(iii)
(i) Cybersecurity Risk Index	**1.000**		
(ii) IVOL	**-0.214** ***	**1.000**	
(iii) ROA	**0.081** ***	**-0.419** ***	**1.000**
Panel B: The Effect of IVOL and ROA	[1]	[2]	[3]
Cybersecurity Risk Index	**0.298** ***	**0.143** ***	**0.202** ***
	[6.28]	[3.83]	[4.85]
IVOL	-	**-0.847** ***	-
		[-12.29]	
ROA	-	-	**0.606** ***
			[10.35]
Observations	409,016	406,850	407,700
Panel C: Orthogonalized Variables	[1]	[2]	[3]
Cybersecurity Risk Index	**0.124** ***	-	-
	[2.80]		
Cybersecurity Risk Index	-	**0.175** **	-
(Orthogonal to IVOL)		[2.33]	
Cybersecurity Risk Index	-	-	**0.164** ***
(Orthogonal to ROA)			[3.36]
Controls	Yes	Yes	Yes
Observations	342,573	342,573	342,573

Table IA.13
Fama MacBeth Regressions: Exposure to Cybersecurity Risk Factor

This table reports the results from Fama-MacBeth regressions on the relation between Cyber Beta (calculated using rolling firm-level regressions of monthly returns on our cybersecurity risk factor over the previous 60 months) and subsequent monthly stock returns (1-month). For each month of our sample, we run cross-sectional regressions of excess stock returns on lagged Cyber Beta and a set of firm characteristics that are also lagged. These include beta, size, book-to-market, momentum, short-term reversal, illiquidity, coskewness, idiosyncratic volatility, asset growth, profitability, R&D, demand for lottery-like stocks (max), length of Item 1A. Risk Factors of the Form 10-K and 10-K readability. All variables are defined in Appendix B. The continuous variables are standardized to have a mean of 0 and standard deviation of 1. The coefficients are reported as time-series averages of the estimates from the cross-sectional regressions. The *t*-statistics, which are reported in brackets, are based on the Newey-West heteroskedasticity and autocorrelation consistent standard errors. * and *** denote statistical significance at 10% and 1% levels, respectively.

	$Returns_{t+1}$
Cyber Beta	**0.099 *****
	[3.42]
Beta	**-0.049**
	[-0.59]
Market Value	**-0.047**
	[-0.42]
Book-to-Market	**0.031**
	[0.66]
Momentum	**0.164 ***
	[1.74]
Reversal	**-0.045**
	[-0.58]
Illiquidity	**0.030**
	[0.67]
CoSkew	**-0.014**
	[-0.31]
Indiosyncratic Volatility	**-0.469 *****
	[-5.62]
Asset Growth	**-0.034**
	[-0.59]
ROA	**0.630 *****
	[8.89]
R&D	**0.386 *****
	[4.16]
Max	**-0.354 *****
	[-3.18]
Risk Section Length (ln)	**-0.032**
	[-0.95]
Readability (ln)	**0.064**
	[0.92]
Constant	**0.519**
	[1.37]
Observations	171,945

Table IA.14
Cybersecurity Risk and Future Cyberattacks: Comparison across Measures

This table reports the results of logit regressions of cybersecurity risk, as measured in three different ways, on future cyberattacks. The dependent variable is a dummy variable that equals 1 if the firm experiences a cyberattack at time *t+1*, and zero otherwise. To measure cybersecurity risk, we firstly use our own measure (Cybersecurity Risk Index) in Model 1 (benchmark specification). In Model 2, we alternatively use the length (number of sentences) of cyber-related disclosures (Cyber-related Disclosures). In Model 3, we use an *ex-ante* measure of cybersecurity risk calculated after using fiscal year *t-1* variables to predict year *t* cyberattacks and then using the coefficients of this regression to construct the cyberattack probability for *t+1* (*Cyberattack Probability*). Model 4 simultaneously controls for all three cybersecurity risk measures. All models include the controls used in Model 2 of Table 6, which are lagged. The continuous variables are standardized to have a mean of 0 and standard deviation of 1. Standard errors are clustered at the firm level. *** indicate statistical significance at the 1% level.

	Model 1	Model 2	Model 3	Model 4
Cybersecurity Risk Index	**0.656 *****	-	-	**0.438 *****
	[4.60]	-	-	[2.82]
Cyber-related Disclosures	-	**0.353 *****	-	**0.248 *****
	-	[5.88]	-	[3.17]
Cyberattack Probability	-	-	**-0.022**	**-0.010**
	-	-	[-0.91]	[-0.29]
Clustered SE	Firm	Firm	Firm	Firm
Industry fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Control Variables	Yes	Yes	Yes	Yes
Number of Observations	30,830	37,820	31,294	30,830
Pseudo-R-squared	0.223	0.223	0.213	0.229